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**EC-505 (GS)**  
**B.E. V Semester** Examination, June 2020  
**Grading System (GS)**  
**Communication Networks and Transmission Lines**  
**Time : Three Hours**

**Maximum Marks : 70**

**Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. a) Differentiate between Image impedance and iterative impedance.  
b) Discuss characteristics preparation co-efficient and Image transfer co-efficient.
2. Design a symmetrical  $600\Omega$  bridged T resistance attenuator to have an attenuation of 20 dB.
3. Distinguish between symmetrical and asymmetrical attenuator.
4. Realize the given impedance function in caver-I form:

$$Z(s) = \frac{4(s^2 + 1)(s^2 + 9)}{5(s^2 + 4)}$$

5. The characteristics impedance of a 805m-long transmission line is  $4 - 23.2^\circ \Omega$  the attenuation constant is  $74.5 \times 10^{-6} \text{ Np/m}$  and the phase shift constant is  $174 \times 10^{-6} \text{ rad/m}$  at 5kHz calculate the line parameters R, L, G and C per meter and the phase velocity on the line.
6. What is propagation constant? Derive the formula of it for the open wire transmission line.

OR

Discuss passive LC filters.

7. a) What is Attenuator? Derive design equations for a T type attenuator.  
b) What is Distortion less line? Explain waveform distortion.
8. Write short notes (any four)
  - a) Smith Chart
  - b) RLC Network synthesis
  - c) Short circuit line
  - d) Coaxial cable
  - e) Micro strip lines

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